

**P. R. Pote Patil College of Engineering and Management,
Amravati**

Title of Project

“Product Inventory & Order Management System”

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CERTIFICATE

This is to certify that following group of students of have successfully completed the DBMS Lab project titled “**Product Inventory & Order Management System**” for the requirements of the Bachelors of Technology (B.Tech) program in Node.js Laboratory during the academic year 2025–26.

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**P. R. Pote Patil College of Engineering & Management, Amravati Session
2025 – 26**

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Chapter 1

INTRODUCTION

1.1 Objective

1. Develop a user-friendly application using Flask, Python, MySQL, HTML, CSS, and JavaScript.
2. Provide secure registration and login facilities for sellers and buyers.
3. Allow sellers to add, update, view, and manage products.
4. Maintain product details such as name, SKU, category, price, quantity, and minimum stock level.
5. Allow buyers to add products to a cart and place orders.
6. Help sellers manage and update order statuses.
7. Track inventory changes and identify products with low stock.
8. Maintain order and inventory records in MySQL.
9. Provide dashboards and reports to monitor business activities.

1.2 Introduction

Inventory and order management are important activities in every business that sells products. Maintaining product details, tracking available stock, processing customer orders, and updating inventory manually can be time-consuming and may lead to errors. The Inventory & Order Management System is a web-based application designed to manage these activities through a single platform. Sellers can manage products and inventory, while buyers can browse products, add items to their shopping cart, and place orders.

The application is developed using Python and Flask for the backend, MySQL for database management, and HTML, CSS, JavaScript, and Bootstrap for the user interface. Flask handles application routes and business logic, while MySQL stores user details, product information, categories, cart items, orders, and inventory transactions.

The system has two main roles. Sellers can manage categories, products, stock, orders, and reports. Buyers can browse products, view details, add items to the cart, place orders, view order history, and manage profile information.

Product Inventory & Order Management System

The system also maintains inventory transaction records to help track stock changes. This makes the application useful for demonstrating the major activities involved in an online store and inventory management process.

1.3 Motivation

The motivation behind this project is to understand how an online shopping and inventory system works in a real business environment. Many small businesses maintain product and stock records manually using registers or spreadsheets. This can make it difficult to track stock availability, manage customer orders, and identify products that need to be restocked.

The project provides an opportunity to develop a practical application that combines product management, inventory control, and order processing.

- Maintain product and customer records in a centralized database.
- Make product information available through an online categories
- Simplify the process of placing and managing orders.
- Understand database connectivity and CRUD operations using Flask and MySQL.
- Gain practical knowledge of web development, authentication, and role-based access.
- Develop a project that can be extended with additional e-commerce features.
- Reduce manual work in product and stock management.

Chapter 2

PROPOSED WORK

2.1 Basic Idea

The basic idea is to create an online inventory and order management platform that connects sellers and buyers. Sellers manage products and stock through a seller dashboard. Each product contains information such as product name, SKU, category, description, price, available quantity, minimum stock level, and product status.

Buyers use the buyer dashboard to browse the product categories search for products, view product details, add products to the cart, and place orders by entering delivery information. Buyers can also view previous orders and check their current status.

The system stores information in a MySQL database. When a buyer places an order, the application records the order and its items. Inventory information is updated according to order and stock-management operations. Sellers can monitor products whose available quantity has reached or fallen below the minimum stock level.

2.2 System Architecture/ Workflow



2.3 Methodology

The methodology for developing the Product Inventory & Order Management System follows a simple and structured approach to create an interactive and user-friendly learning system.

Development

1. Frontend:

- **HTML, CSS, JavaScript** – Used to design a responsive and interactive user interface.

2. Backend:

- **Python (Flask)** – Used to develop lightweight REST APIs, handle application logic, and manage communication between the frontend and backend.

3. Database:

- **MySQL** – Used to store and manage employee information. It provides efficient data storage, retrieval, and integration with the backend through a database server.

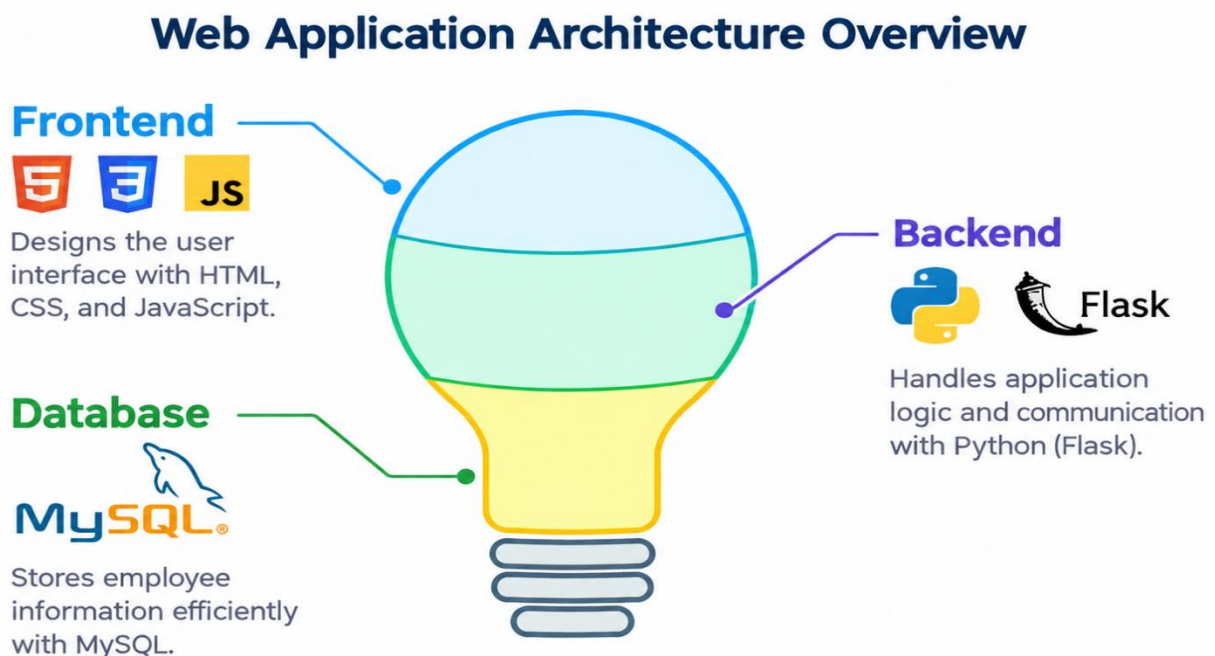


Figure 2.3.1: Methodology

Chapter 3

RESULT

3.1 Result

The Inventory & Order Management System is designed to provide a centralized platform for managing products, inventory, and customer orders. The application includes separate seller and buyer interfaces, allowing each type of user to access the required features.

The seller can manage products and categories, monitor stock levels, view orders, and access reports. The buyer can browse products, add items to the cart, place orders, and view order information.

The application uses Flask for backend processing and MySQL for storing application data. The modular structure separates authentication, product management, inventory, cart, and order operations.

3.2 Demonstration

InventoryHub Shop My Orders 🛒 Harshal ▾

Hi Harshal, welcome back!

 **1**
TOTAL ORDERS

 **0**
IN PROGRESS

 **1**
DELIVERED

Recent Orders [View all](#)

Order	Date	Amount	Status	
ORD-2026-249237	25 Sep 2026	₹91,097.00	Delivered	View

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Figure 3.2.1: Buyer Dashboard

Product Inventory & Order Management System

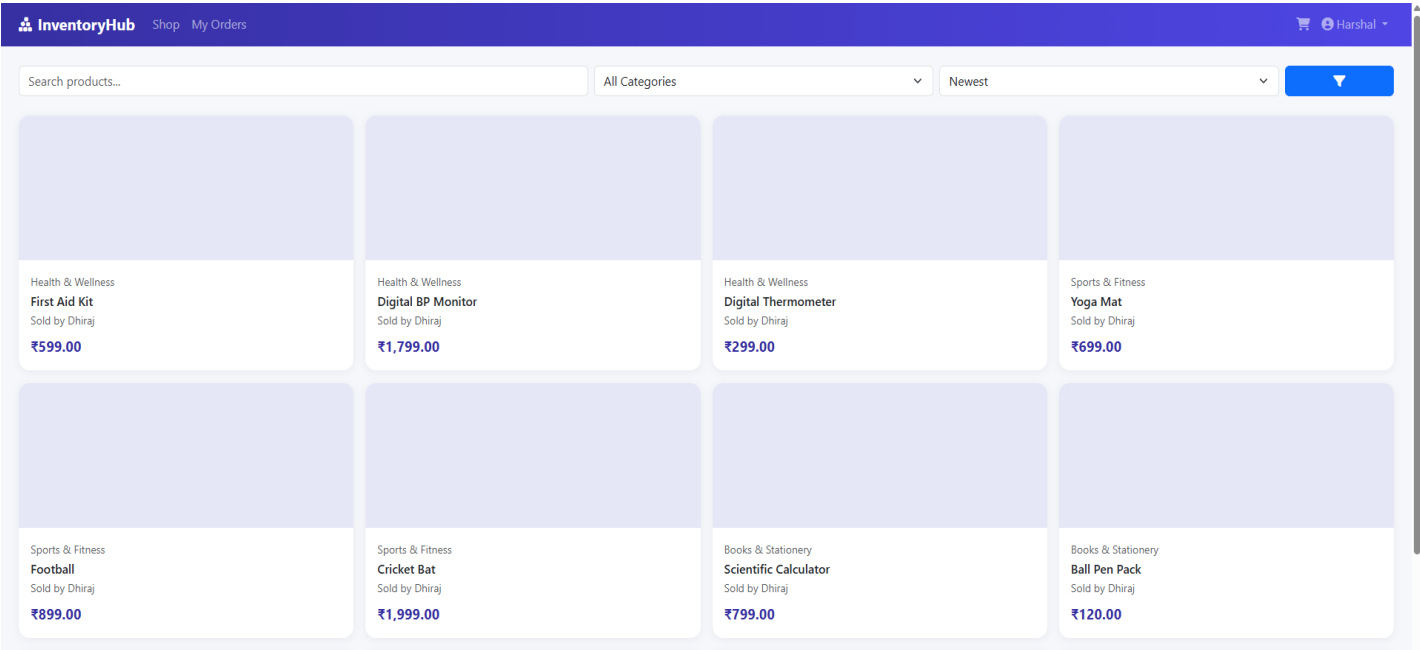
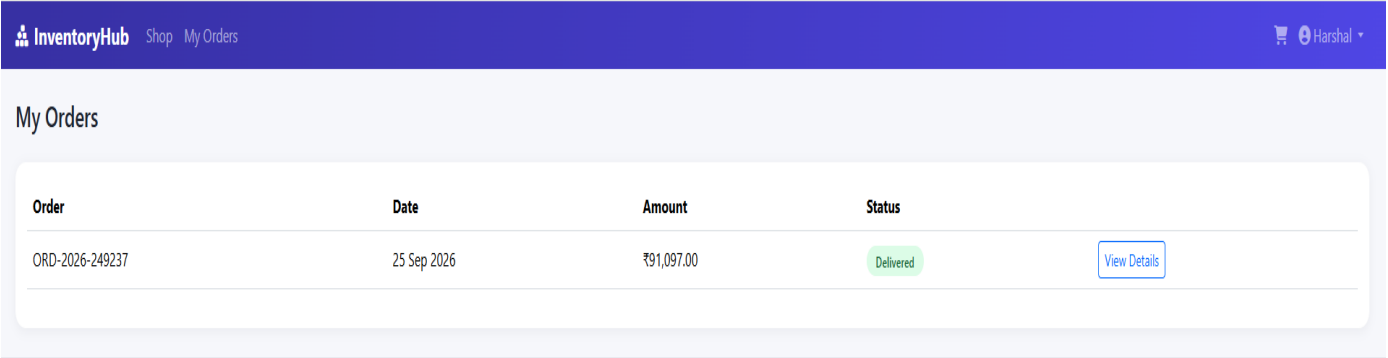
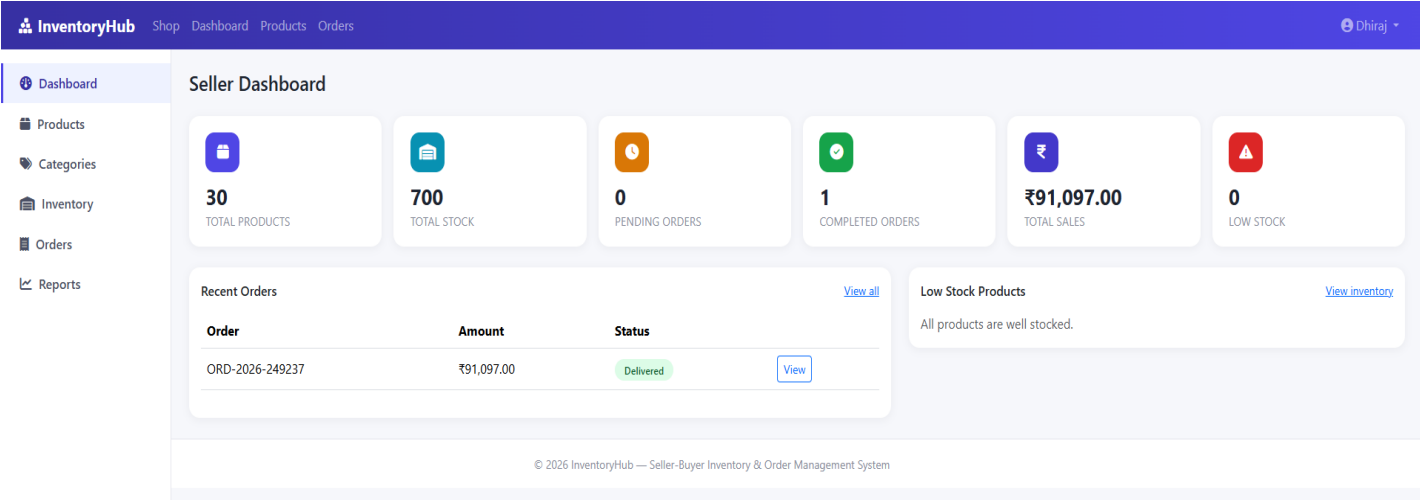


Figure 3.2.2: Shopping Page



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Figure 3.2.3: My-order Page



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Figure 3.2.4: Seller Dashboard

Product Inventory & Order Management System

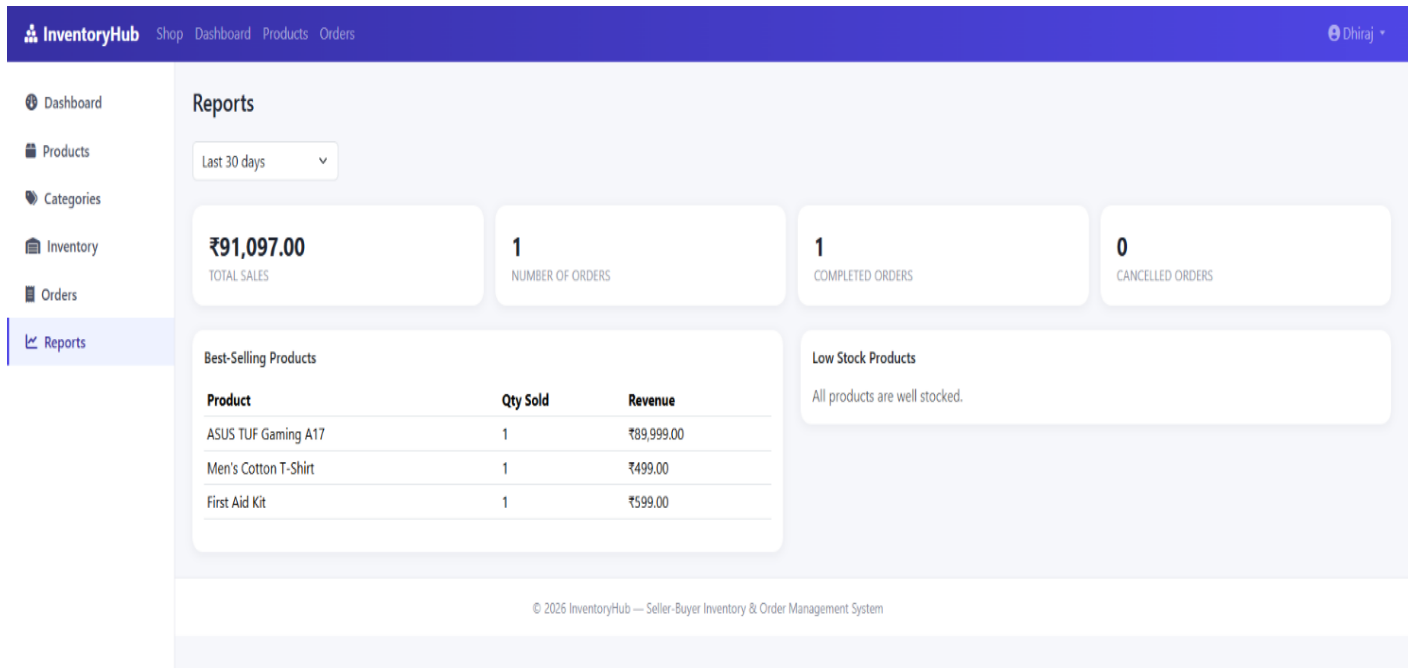


Figure 3.2.5: Analysis Page

InventoryHub	Shop	Dashboard	Products	Orders	Dhiraj
Dashboard	Products	Categories	Inventory	Orders	Reports
Inventory					
Current Stock					
Product	SKU	Stock	Min. Level	Status	Adjust
ASUS TUF Gaming A17	ELE-ASU-001	7	2	OK	+/- Reason (optional) Apply
LG 260L Refrigerator	APP-LG-001	7	2	OK	+/- Reason (optional) Apply
Samsung Washing Machine	APP-SAM-002	8	2	OK	+/- Reason (optional) Apply
iPhone 13	MOB-APP-003	10	3	OK	+/- Reason (optional) Apply
Office Chair	HOM-CHA-002	10	3	OK	+/- Reason (optional) Apply
HP Pavilion 15	ELE-HP-002	12	4	OK	+/- Reason (optional) Apply
Study Table	HOM-TAB-001	12	3	OK	+/- Reason (optional) Apply
Lenovo IdeaPad Slim 3	ELE-LEN-003	15	5	OK	+/- Reason (optional) Apply
Bookshelf	HOM-BOO-003	15	3	OK	+/- Reason (optional) Apply
Cricket Bat	SPO-BAT-001	15	3	OK	+/- Reason (optional) Apply

Figure 3.2.6: Inventory Page

Chapter 4

CONCLUSION & Reference

4.1 Conclusion

The Inventory & Order Management System is a web-based application developed to manage product information, inventory, and customer orders. It provides separate interfaces for sellers and buyers, allowing them to perform their respective activities through a common platform.

The system uses Flask and Python for backend development, MySQL for database management, and HTML, CSS, JavaScript, and Bootstrap for the user interface. The seller module supports product and category management, inventory monitoring, order management, and reports. The buyer module supports product browsing, cart management, checkout, and order history.

The project demonstrates the practical use of web development concepts, database connectivity, CRUD operations, user authentication, role-based access, and inventory management. Overall, the project provides a foundation for an online shopping and inventory system. It can be extended with additional features to support more advanced business requirements.

4.2 Reference

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